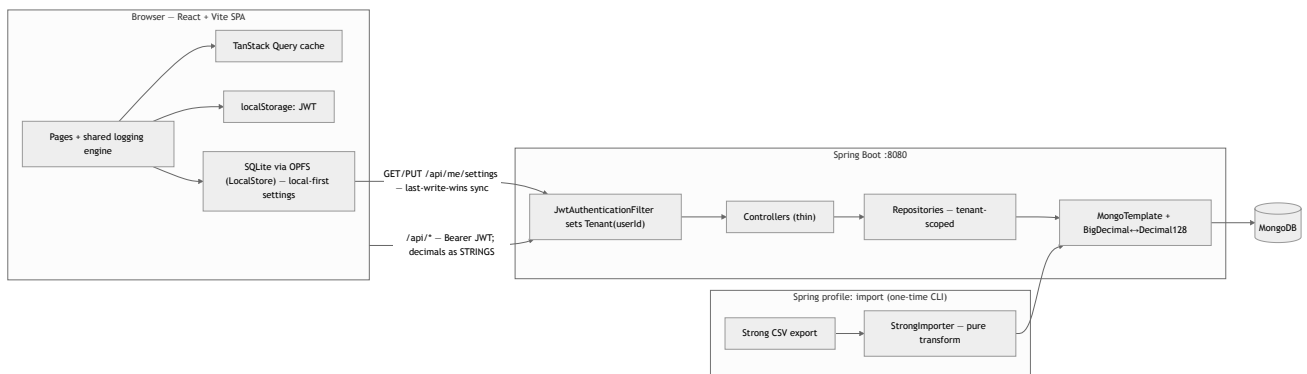


Workout Logger — diagrams

Structural and behavioural diagrams (Mermaid, renders on GitHub). See `DESIGN.md` for the authoritative architecture record and `CLAUDE.md` for invariants. Diagrams reflect the current code.

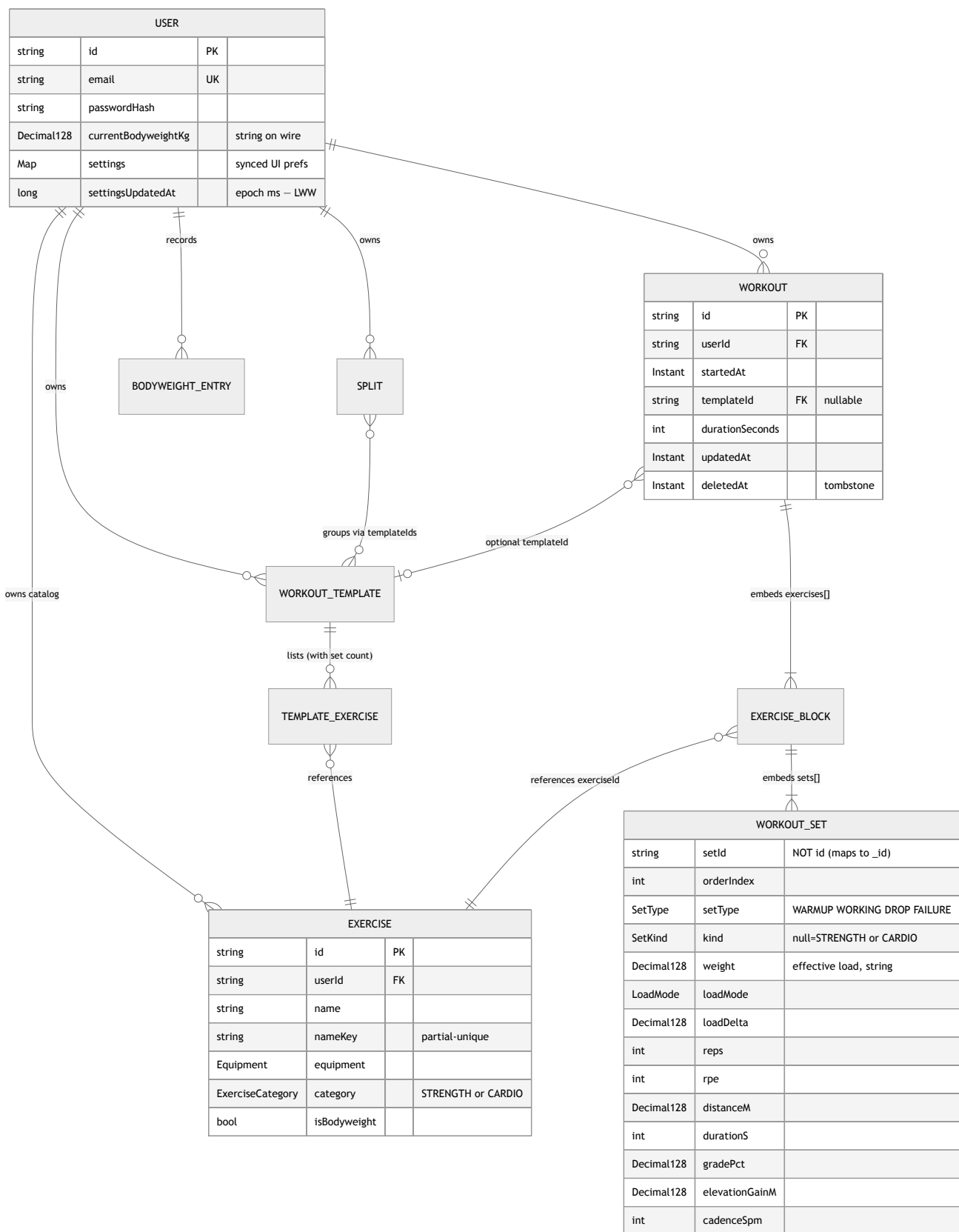
Structural

1. System architecture (components & runtime)

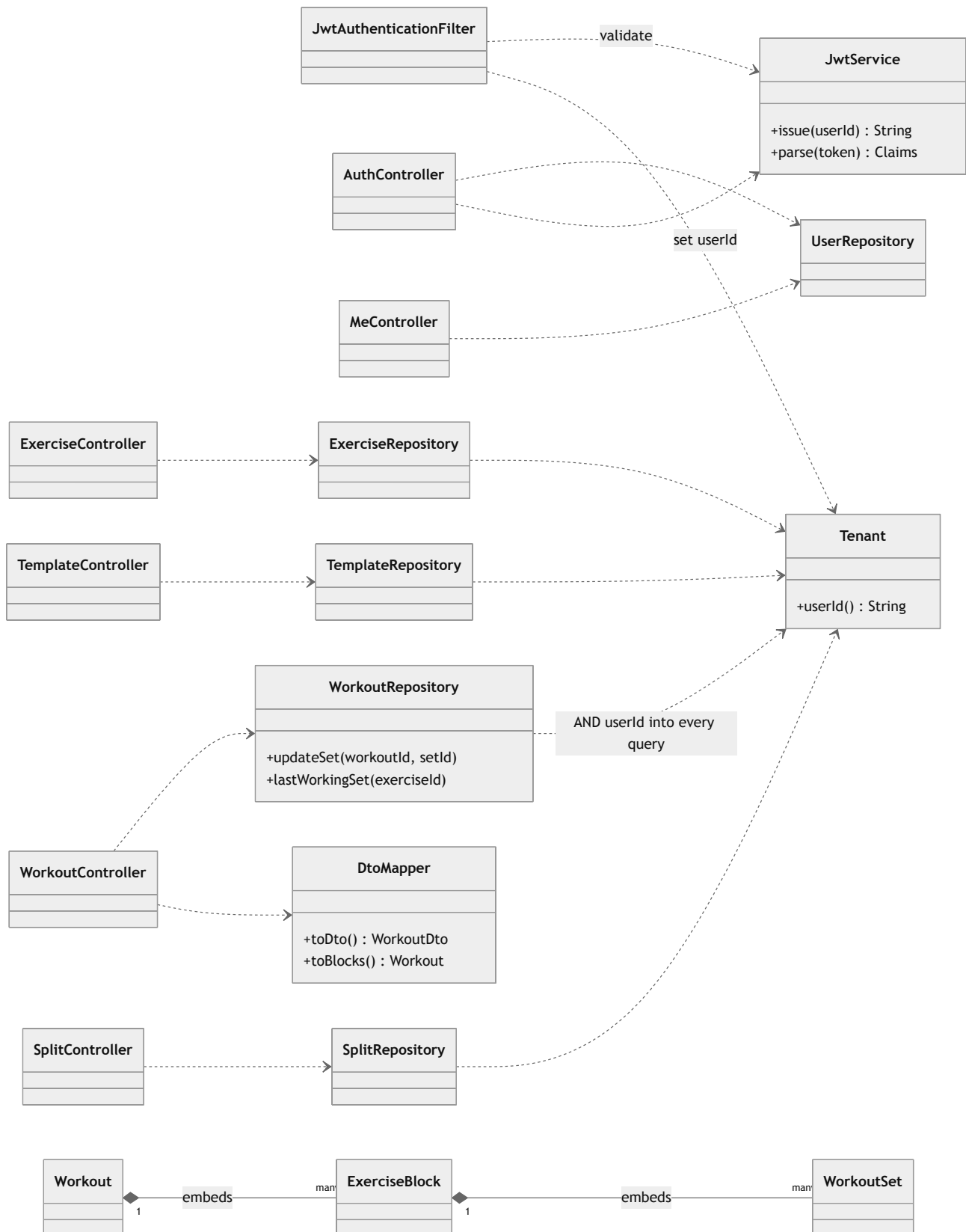


2. Data model (collections, embedding & references)

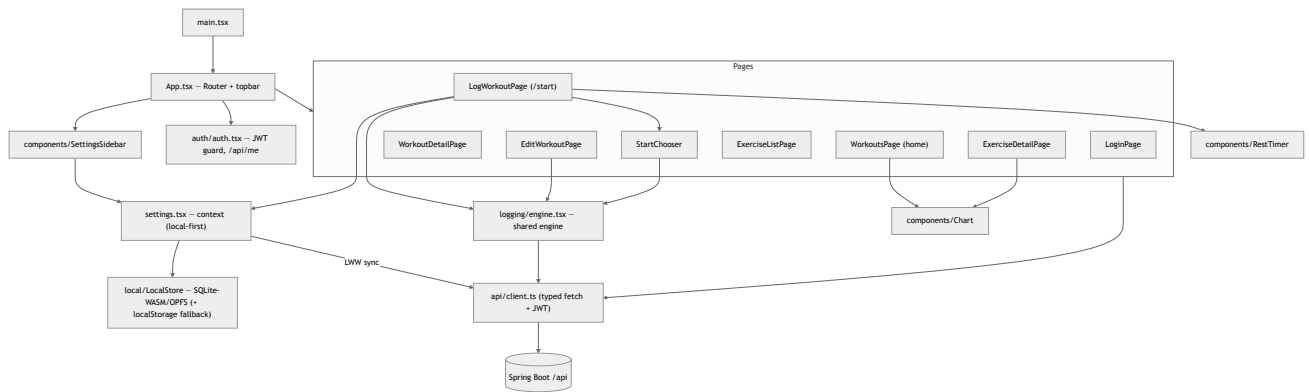
A **workout session** is one document that embeds `exercises[]`, each embedding `sets[]`. Everything else relates by id reference (many-to-many), not joins. Every collection is scoped by `userId`.



3. Backend layers (request path + tenant isolation)

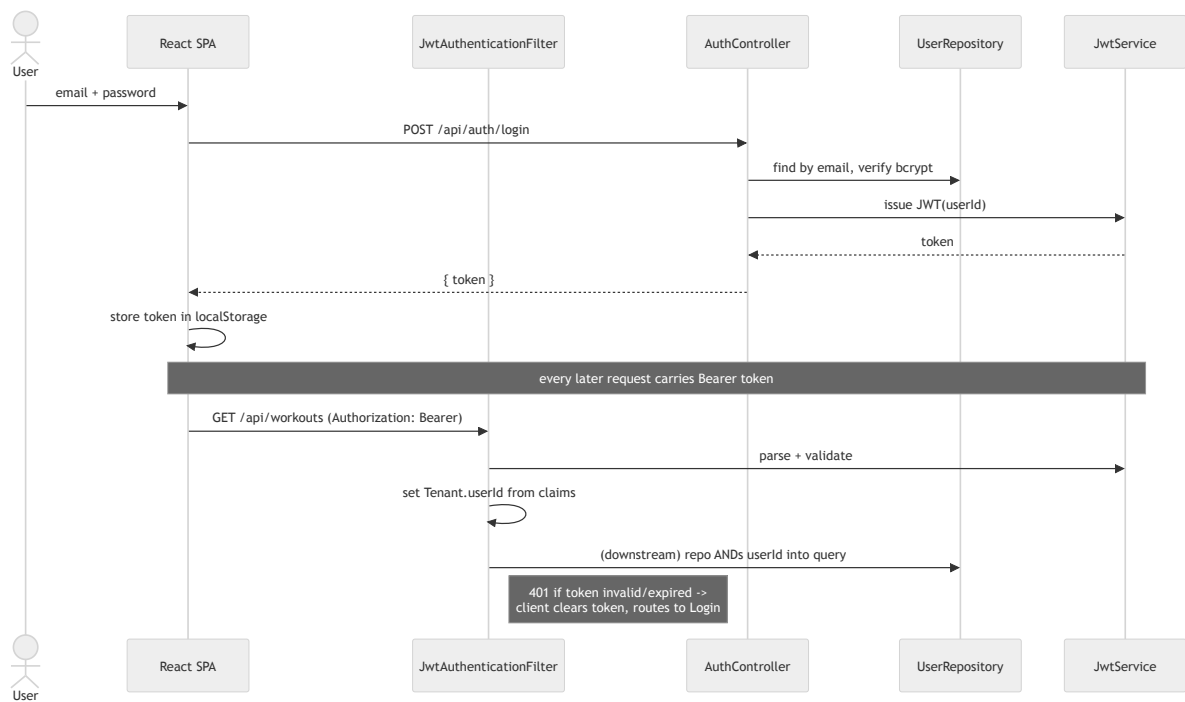


4. Frontend modules

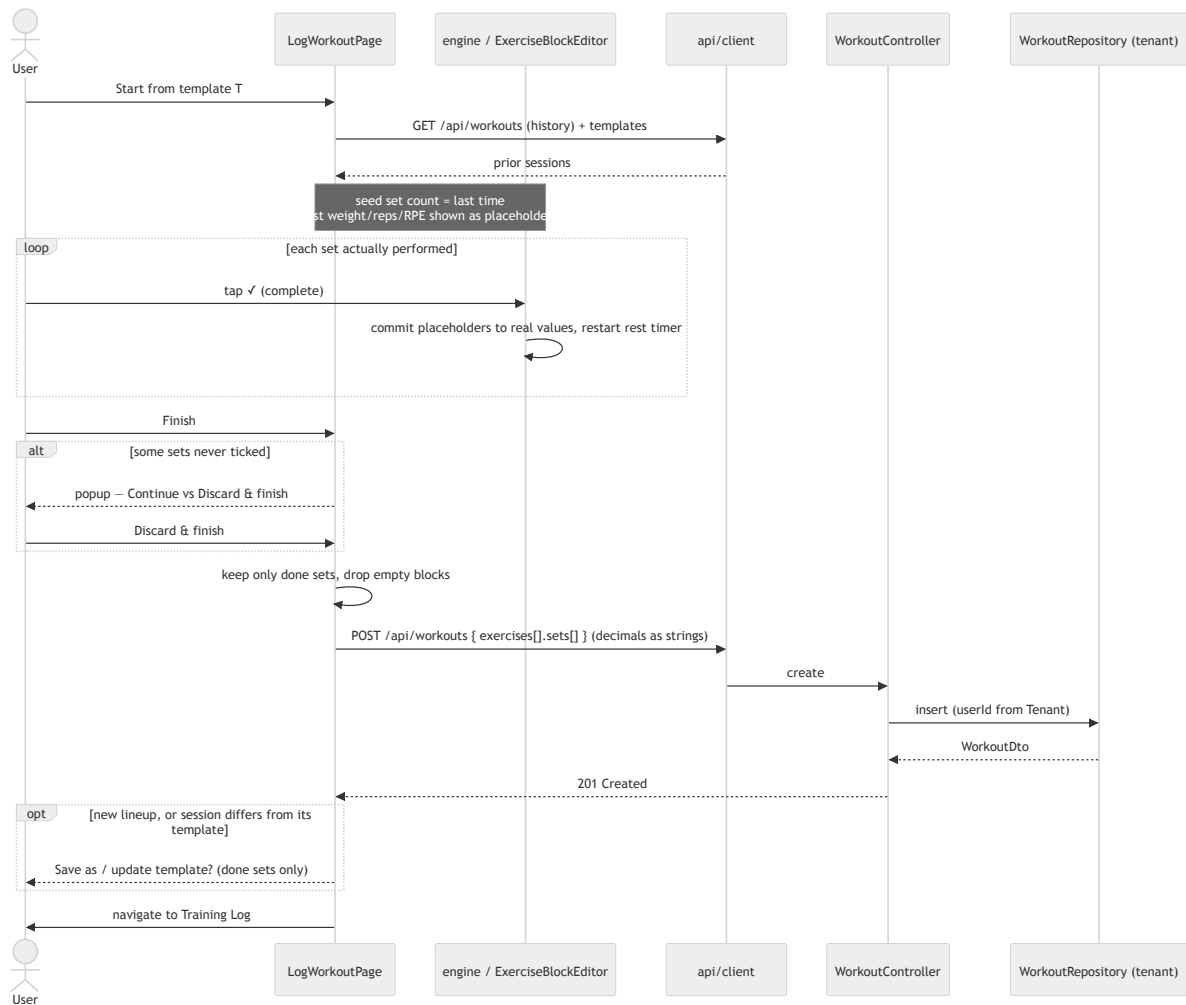


Behavioural

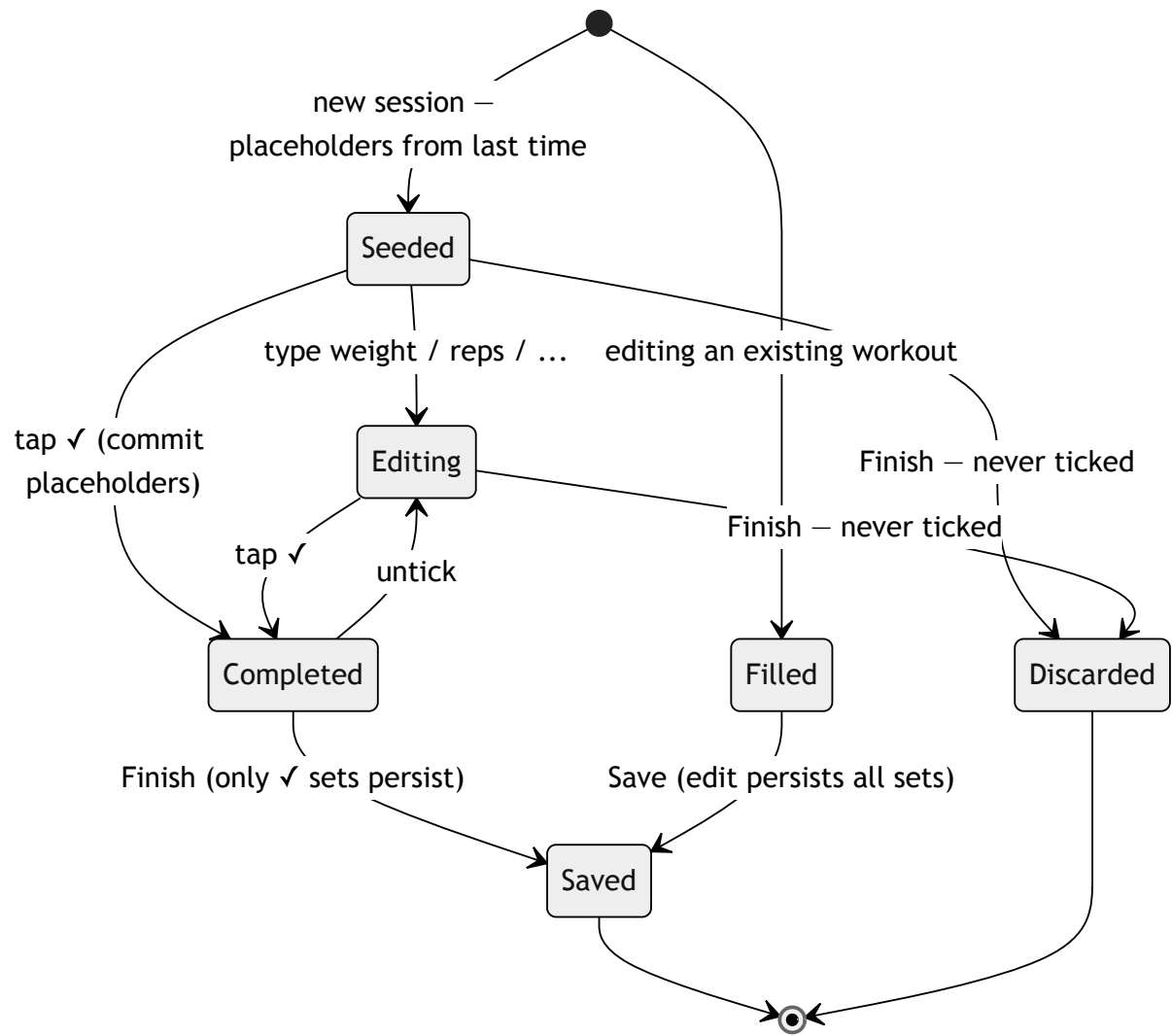
5. Authentication & per-request tenant isolation



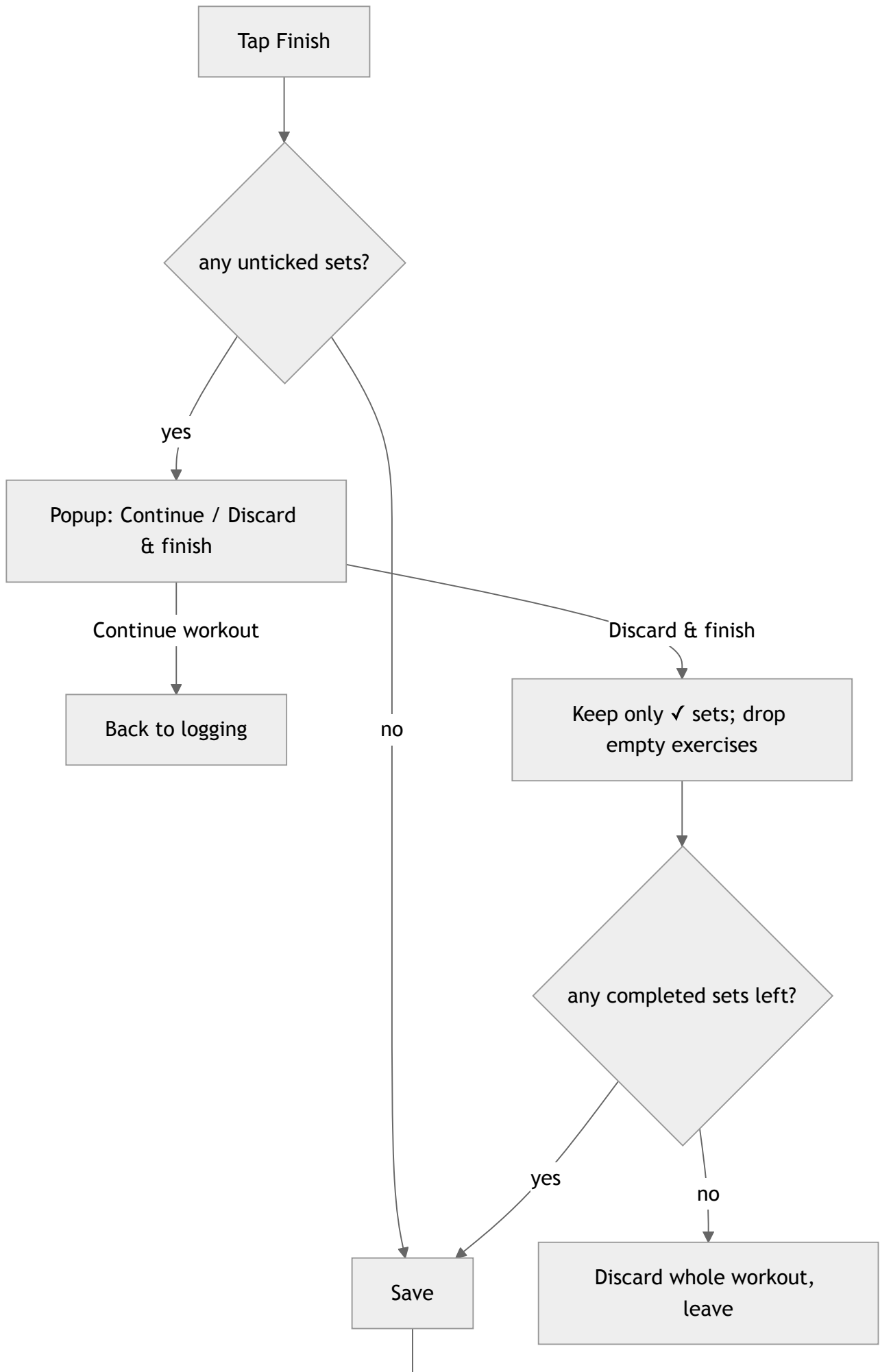
6. Logging a workout (start → complete → finish → save)

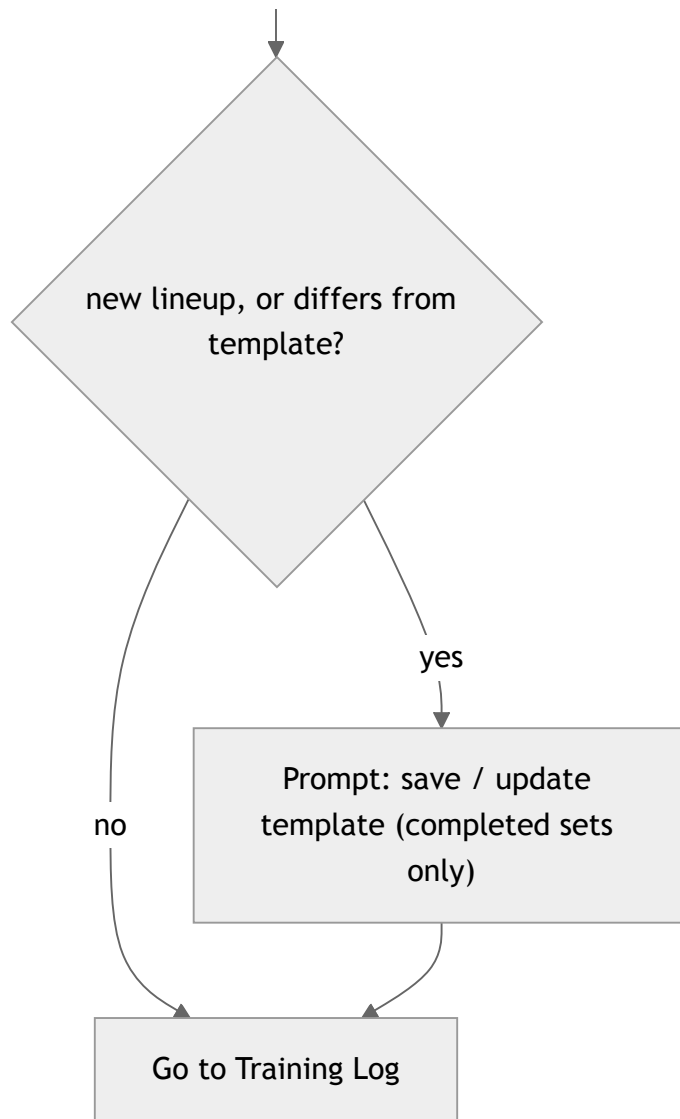


7. A set's lifecycle (placeholder → completed → saved/discarded)

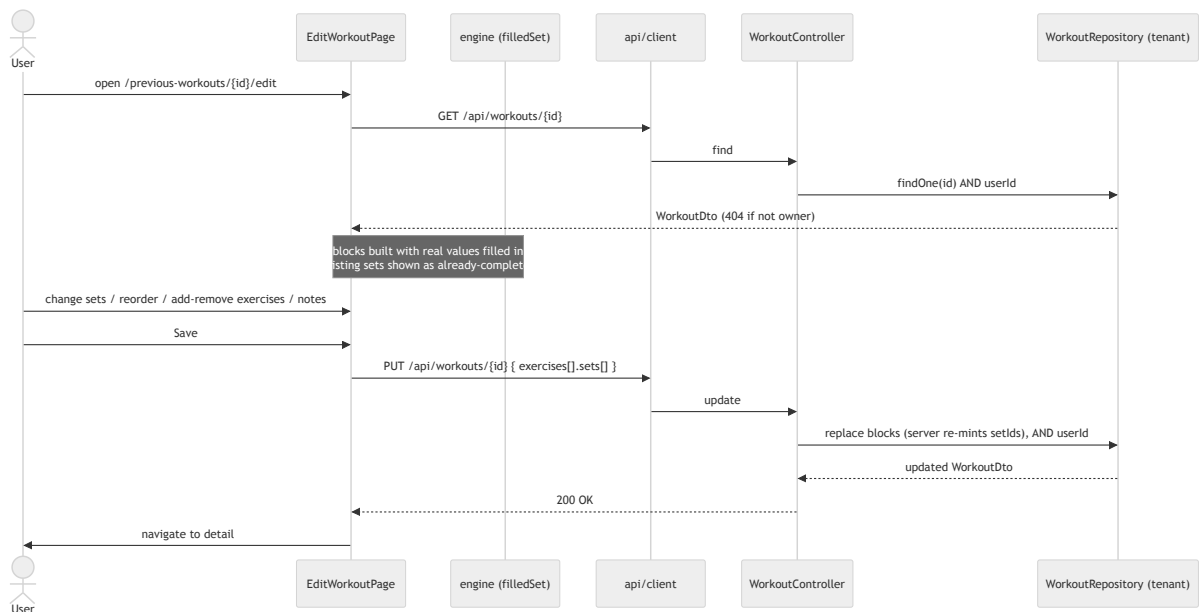


8. Finish-workout decision (discarding unfinished sets)

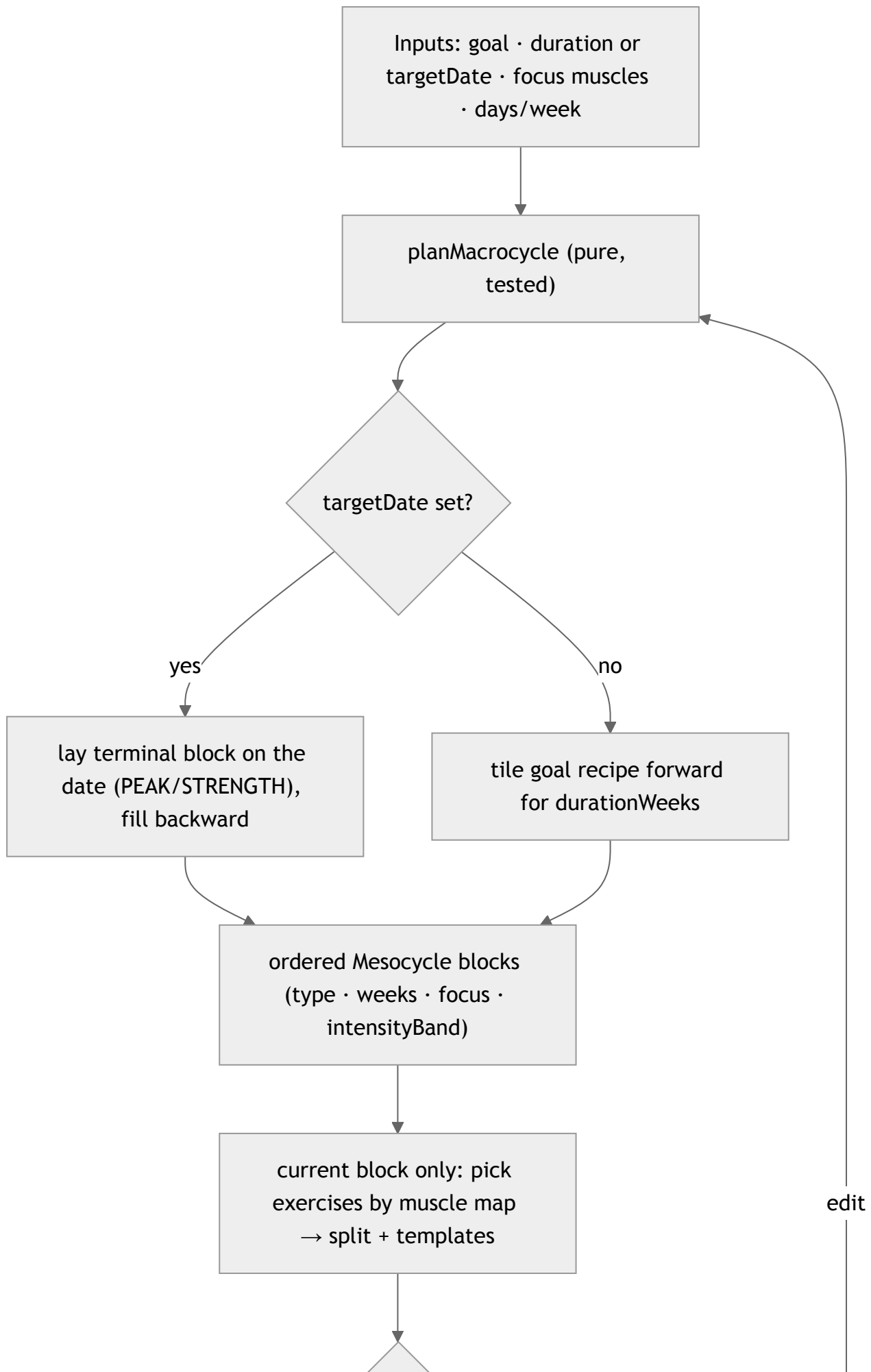


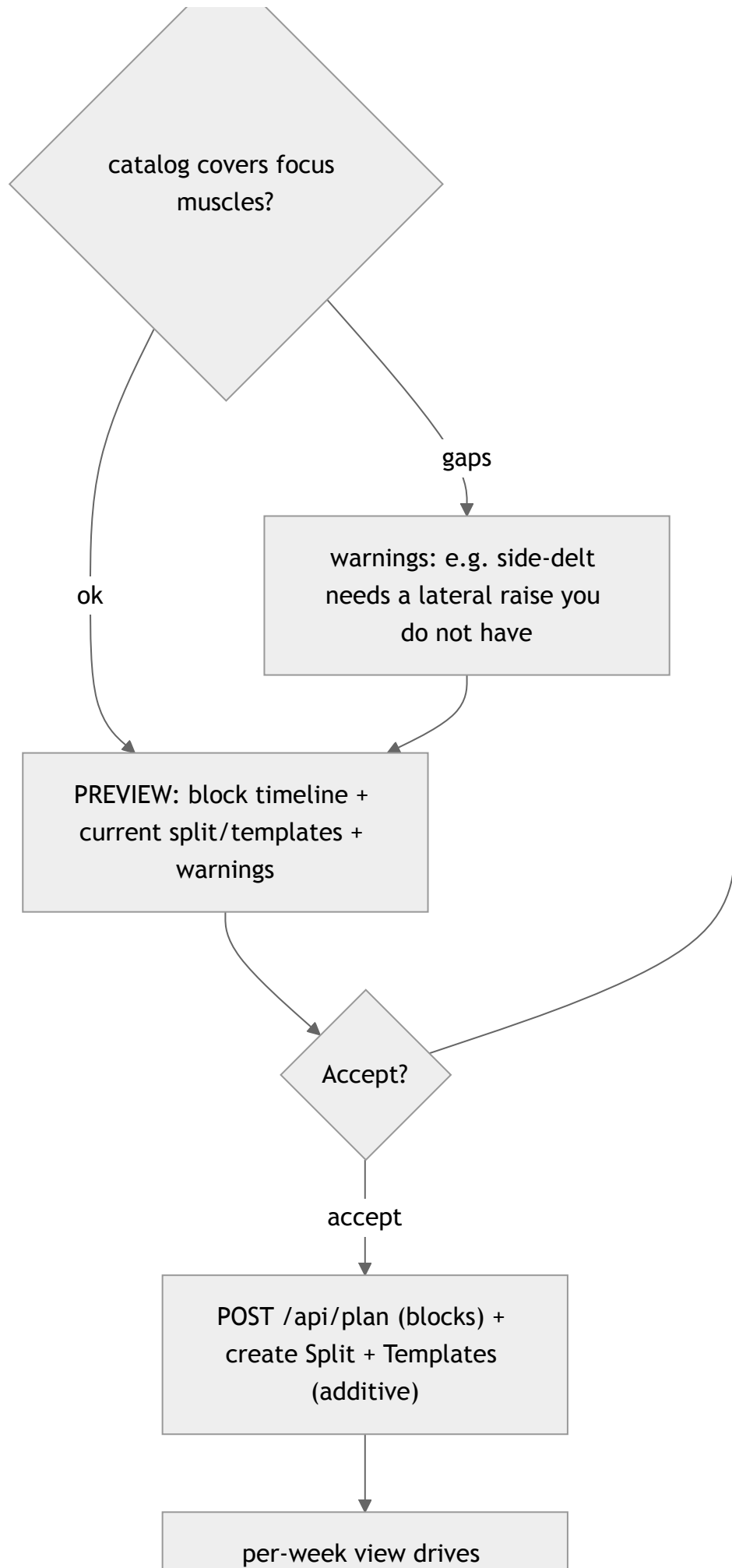


9. Editing a completed workout



10. Macrocycle planner (Layer 4)

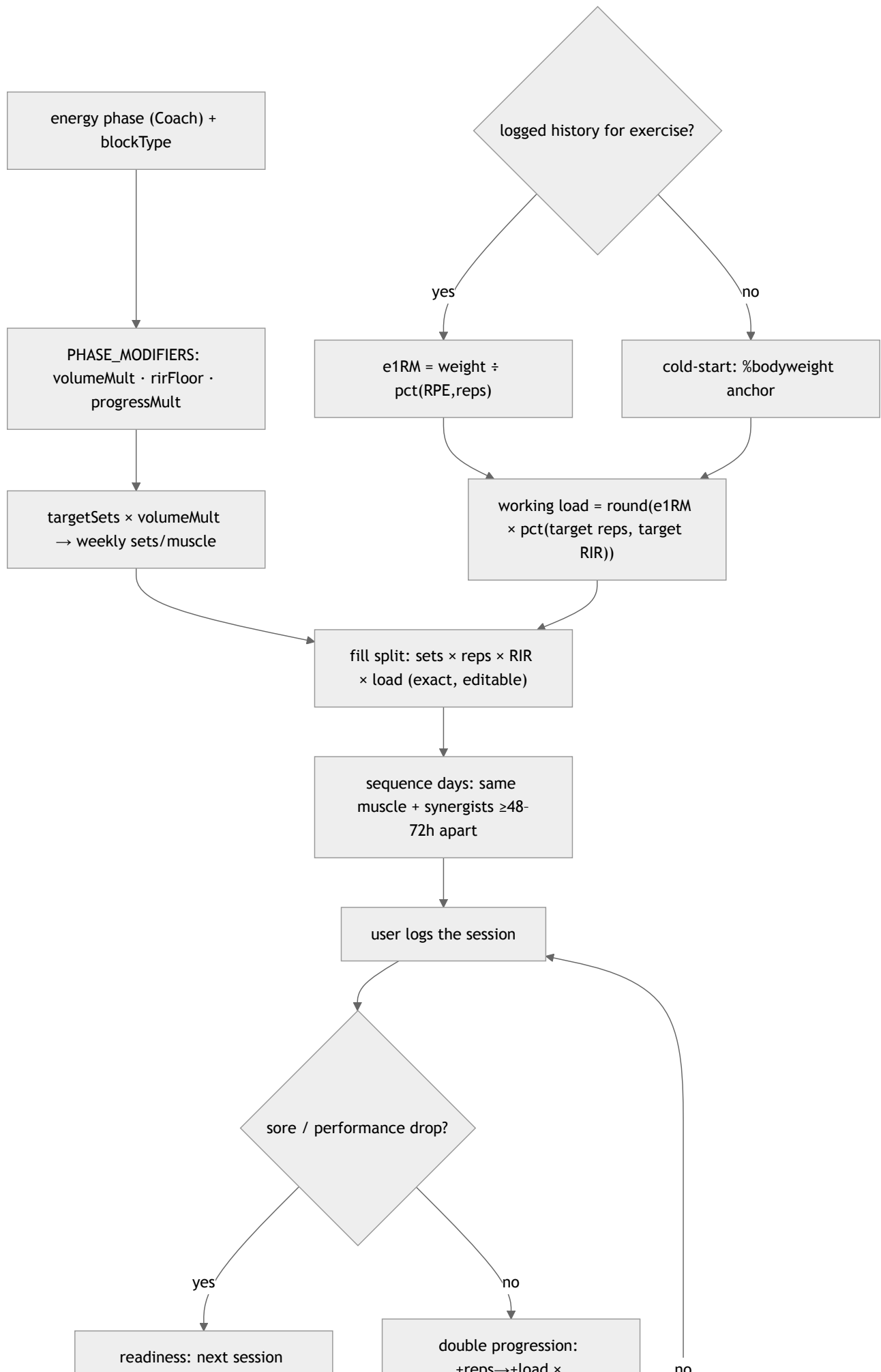


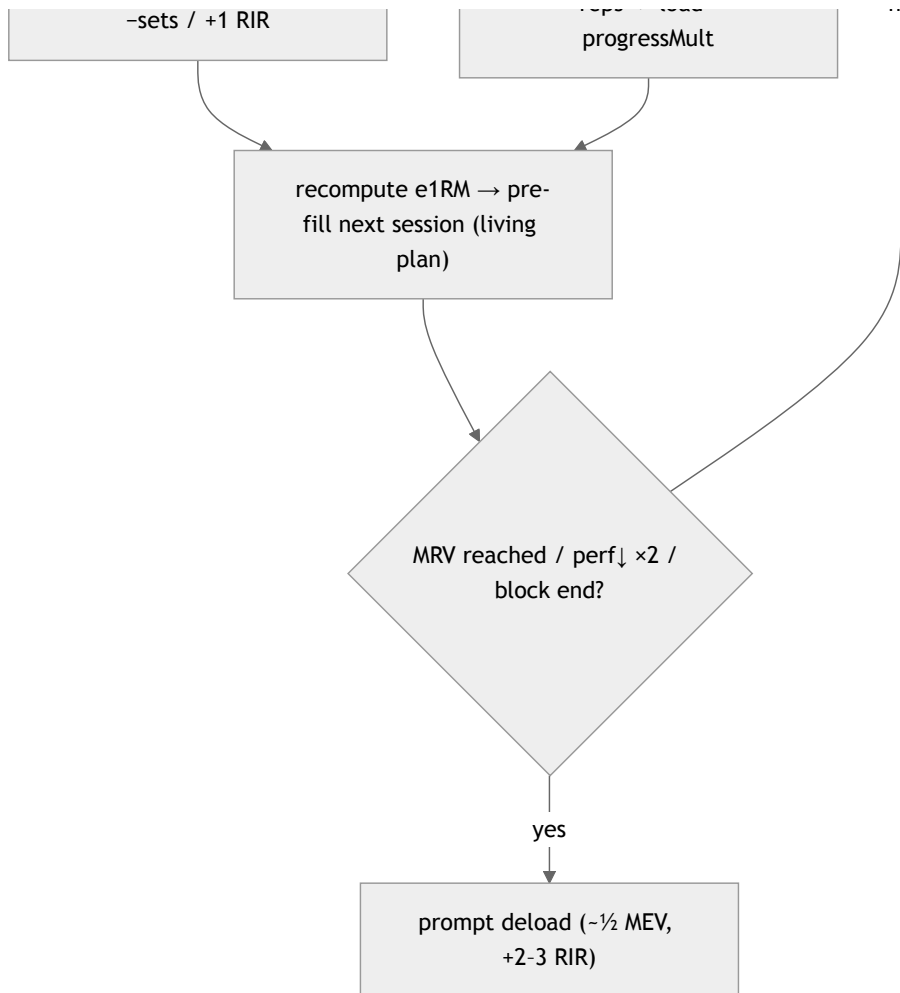


logging; advance through
microcycles

blockType (volume band + reps) and energy phase (deficit-trim) are orthogonal axes; accept creates, never mutates; only the current block's training is materialized — distal blocks stay as intent.

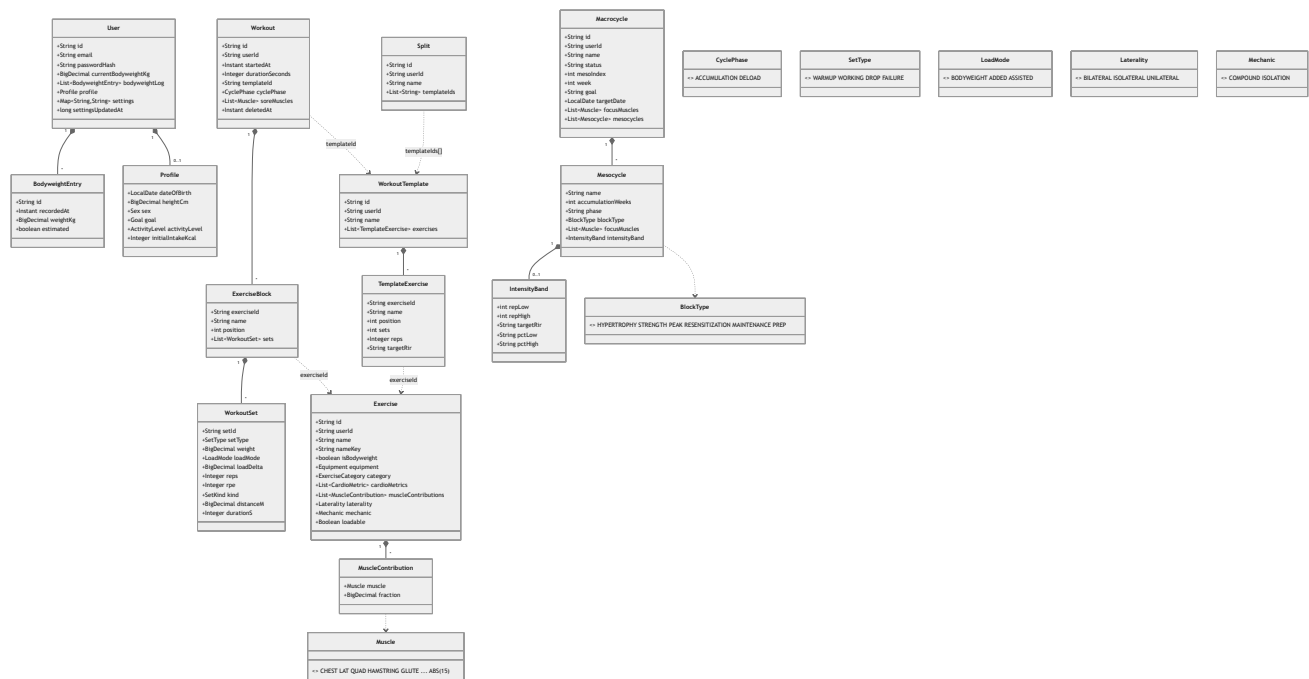
11. Prescription, recovery & autoregulation (Layer 5)





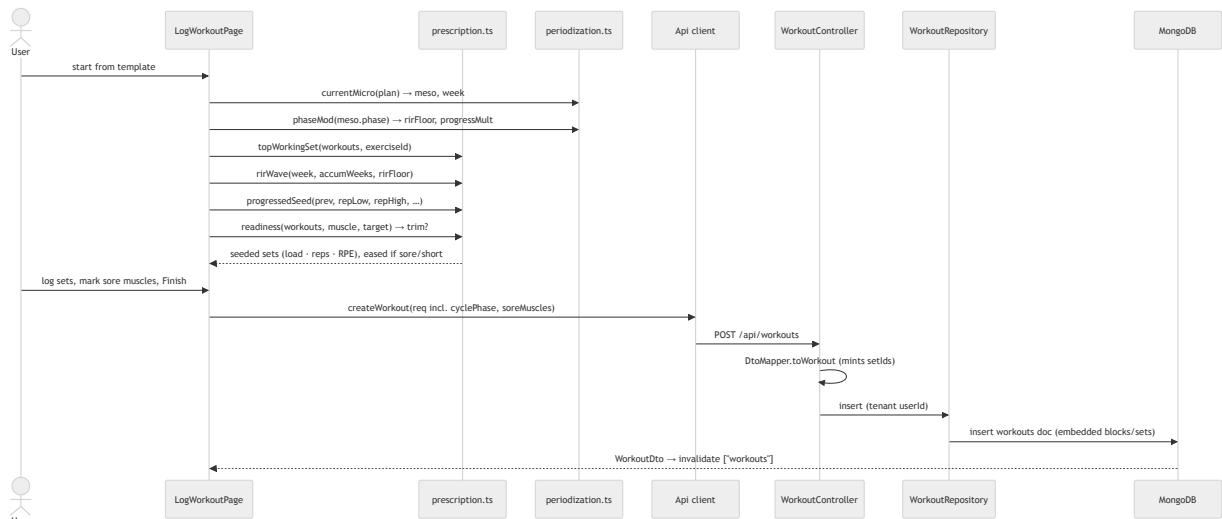
Energy phase scales volume/intensity/progression; numbers seed from logged e1RM (else %BW cold-start); recovery spacing + readiness keep a muscle from being trained fatigued; everything stays an editable preview.

12. Domain model — class diagram (structural)

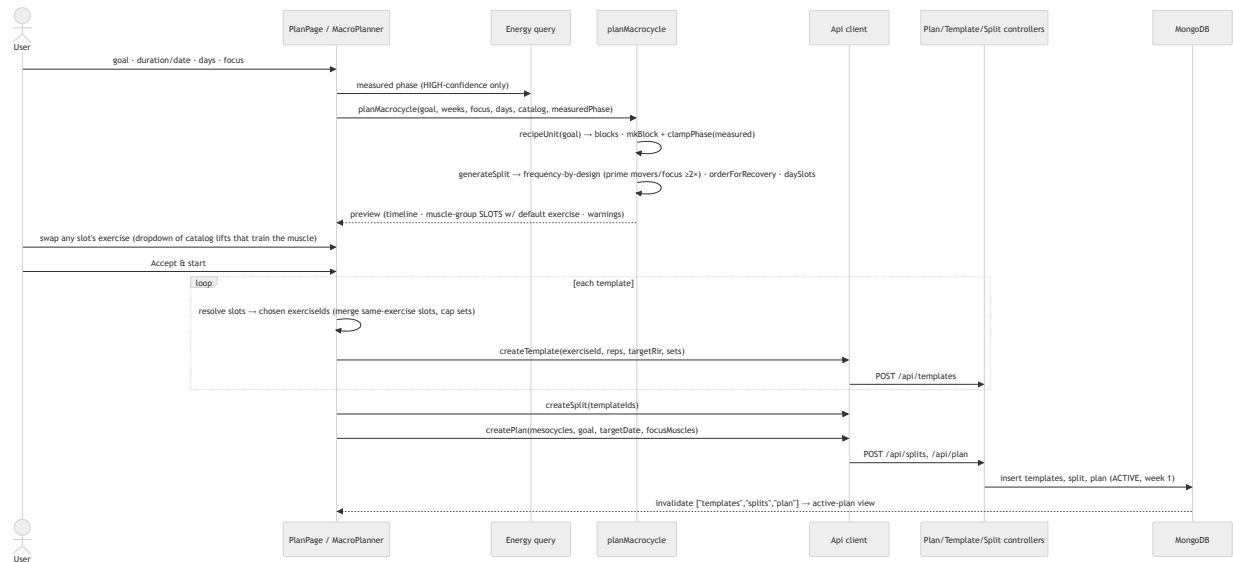


Every aggregate carries `userId` (tenant isolation — every repo ANDs it). Decimals are `BigDecimal` (Decimal128 in Mongo, strings on the wire). The embedded set id is `setId`, not `id`. Also-present enums: `Equipment`, `ExerciseCategory`, `SetKind`, `CardioMetric`, `Sex`, `Goal`, `ActivityLevel`.

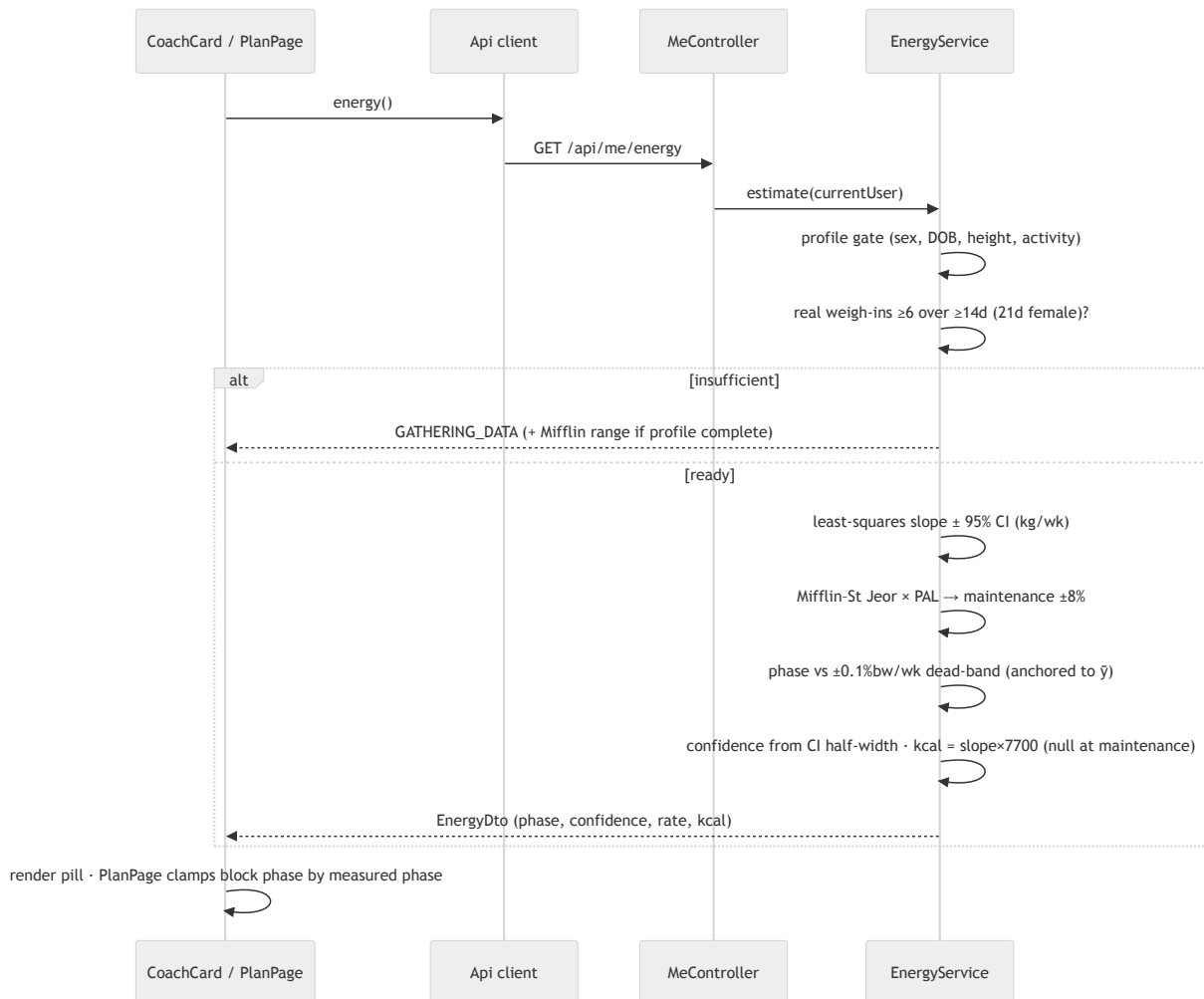
13. Sequence — log a planned session (the living plan)



14. Sequence — build & accept a macrocycle plan



15. Sequence — energy "Coach" estimate (read-time, gated)



16. Sequence — registration → default-catalog seeding

